

Task-Specific Computational Advantage of the *Drosophila* Connectome: A Null-Model-Controlled Study

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Abstract

Does the wiring topology of a real biological neural network – who is connected to whom, and with what synaptic weight – confer a computational advantage over a randomly rewired network with the exact same size and degree distribution? We test this question using the Fly-Wire FAFB v783 connectome of the adult female *Drosophila melanogaster* brain, embedded as a fixed recurrent substrate in a differentiable, Dale’s-law-preserving leaky-rate neural network (**RateBrain**), trained by gradient descent on a variety of computational tasks and compared against a family of null models that progressively relax structural constraints (degree sequence, modularity, weight identity). We find that the real connectome provides a statistically robust advantage (Cliff’s $\delta = 0.61$ – 1.00 , $n = 30$, $p < 0.05$ after Bonferroni-style multiple-testing discipline via a pre-registered magnitude-triggered extension rule) on a specific task format: single-shot, feed-forward, continuous spatial/physical direction regression. This holds across six independent physical/engineering design problems spanning distinct failure modes (geotechnical slip, bending stress, transverse shear, elastic buckling, deflection serviceability, and thin-wall hoop stress), across an order-of-magnitude range of subgraph sizes, and across a $50\times$ range of input sensing scale. The advantage is largely (but not fully) explained by the network’s modular organisation, does not generalise to abstract optimisation, discrete classification, closed-loop control, a different species (*C. elegans*), a different individual of the same species (male *Drosophila* CNS), input noise, or weight quantisation, and does not transfer zero-shot to a mechanically different task in the same family. We report the full boundary-mapping process transparently, including two methodological errors we made and corrected during the study (a ratio-metric artifact and two term-imbalance calibration bugs in newly designed benchmarks), as a case study in the discipline required for this kind of null-model-controlled connectome research.

1 Introduction

A recurring question in computational neuroscience and connectome-based reservoir computing [Suárez et al., 2024, Costi et al., 2025] is whether the specific topology of a mapped biological neural circuit does any real computational work beyond what is explained by simpler statistical properties of the network (size, density, degree distribution). Most prior work in this space reports a positive result against a single, often permissive, null model and does not systematically probe the boundaries of the finding. Our contribution is methodological as much as empirical: we test one connectome, with one consistent statistical framework, across a deliberately wide spectrum of task types, and report negative and positive results with equal rigor, including corrections to our own errors made mid-study.

Our central hypothesis (task–architecture matching) is that any advantage of the real connectome over a randomised network should be largest for tasks structurally close to what the connectome evolved to do – filtering a large, dense sensory input down to a small number of motor outputs – and should weaken or vanish for tasks further from that structural niche.

2 Methods

2.1 Data

The primary connectome is FlyWire FAFB v783 [Dorkenwald et al., 2024], the electron-microscopy-reconstructed connectome of an adult female *Drosophila melanogaster* brain (139,255 neurons, 15,091,983 directed synaptic connections), with cell-type annotations from Schlegel et al. [2024]. Comparison connectomes are the *C. elegans* hermaphrodite somatic nervous system [Cook et al., 2019] and a male *Drosophila* CNS connectome obtained from an independent third-party dataset.

2.2 Subgraph construction

From the full connectome, we select encode (afferent-pool) and decode (efferent-pool) neurons verified, via multi-hop breadth-first search on the full directed graph, to have a genuine forward path between them (a naive random afferent/efferent draw can have zero true connectivity). From this seed set we grow an induced subgraph of a target size (the official size is 3,000 nodes; robustness across 1,000–10,000 nodes is reported in Section 3.3) via breadth-first search along real directed edges.

2.3 Substrate and training

RateBrain is a differentiable, leaky-rate recurrent network with Dale’s-law-preserving weights ($w = \text{sign} \cdot \text{softplus}(\text{gain})$), trained by standard backpropagation (Adam optimiser). For single-shot tasks, sensory input (finite-difference “rays” – local probes of a task’s objective function along each design-variable axis) is injected into encode neurons and held constant for $T=8$ recurrent sub-steps; a linear readout from decode neurons produces a predicted direction. The teacher signal is a crude finite-difference negative-gradient estimate from the same probes (not a hand-crafted heuristic), and training minimises mean squared error against it.

2.4 Null model family

We compare the real connectome against six null models of increasing structural fidelity:

- `er_null` – node/edge count only;
- `scale_free_null` – generic heterogeneous degree distribution;
- `scale_free_correlated_null` – plus in/out-degree correlation;
- `weight_shuffle` – full topology, shuffled weights;
- `community_preserving_rewire` – full degree sequence plus full modularity/community structure;
- `degree_preserving_rewire` – exact degree sequence via Maslov–Sneppen double-edge-swap rewiring (our primary/official null model throughout).

2.5 Statistics

We use the triad of Wilcoxon signed-rank (paired), Mann–Whitney U (unpaired), and Cliff’s δ (effect size) throughout. The official gate is $p < 0.05$ and $|\delta| > 0.33$. The standard protocol is an $n = 8$ pilot; whenever $|\delta| > 0.33$ at $n = 8$, the test is extended to $n = 30$, *independent of whether the pilot p -value was already significant* – a magnitude-triggered, not significance-triggered, extension rule, decided in advance and applied uniformly.

3 Results

3.1 The official finding

Two independent physical/engineering tasks – slope-stability (geotechnical limit-equilibrium factor of safety) and cantilever beam design (bending stress) – show a strong, reproducible advantage of the real connectome over the `degree_preserving_rewire` null at $n = 30$: $\delta = 0.884$ ($p < 10^{-4}$) and $\delta = 0.613$ ($p < 10^{-4}$) respectively. The advantage replicates in 9 of 11 independent subgraph seeds (median $\delta = 0.500$), so it is not an artefact of one fortunate subgraph draw.

3.2 Mechanism

Sweeping the null-model family for the slope-stability task (Table 1) shows that δ falls from 0.884 (degree-preserving) to 0.429 once the null is also constrained to match the real network’s community/modularity structure (`community_preserving_rewire`), and becomes non-significant once weights alone are shuffled on the real topology (`weight_shuffle`, $\delta = 0.258$, $p = 0.088$). Modularity thus explains roughly half of the effect; the remainder is unexplained (a weak, uncorrected exploratory correlation with the fraction of *ascending*-type interneurons was noted but not confirmed).

Table 1: Cliff’s δ across the null-model family (slope stability, $n = 30$).

Null model	Preserves	δ	p
<code>degree_preserving</code>	full degree sequence	0.884	≈ 0
<code>scale_free</code>	generic heterogeneity	0.738	≈ 0
<code>scale_free_correlated</code>	+ degree correlation	0.604	0.00006
<code>community_preserving</code>	+ full modularity	0.429	0.0044
<code>weight_shuffle</code>	topology, no weights	0.258	0.088 (n.s.)

3.3 Generality across the physical-task family

We extended the official pair of tasks to a family of six mechanically distinct single-shot physical/engineering design problems, each with its own failure/design criterion (Table 2). All six show a statistically robust advantage at $n = 30$, with zero exceptions.

Two of the six tasks (column buckling and pressure vessel) initially returned an apparent FAIL and were provisionally reported as such before we discovered, via a suspicious numerical similarity between two independently designed benchmarks’ diagnostic output, that both suffered from an objective-term-balance calibration bug: in one case the material-cost term dominated $> 99.99\%$ of the objective at typical sampled points (making the task trivially solvable by any network regardless of structure), and in the mirror-image case a physical stress term dominated $> 97\%$. Both were

Table 2: Physical/engineering task family, all against `degree_preserving_rewire`.

Task	Failure/design criterion	δ ($n=30$)	p
Beam deflection	serviceability (stiffness)	1.000	3×10^{-11}
Slope stability	geotechnical limit-equilibrium	0.884	≈ 0
Pressure vessel	thin-wall hoop stress	0.871	7×10^{-9}
Column buckling	elastic instability	0.88	5×10^{-9}
Beam shear	transverse shear stress	0.811	7×10^{-8}
Beam bending	bending stress	0.613	≈ 0

corrected (narrowing the design-variable box and/or re-deriving load magnitude as a fraction of a reference cross-section’s capacity, verified numerically before any further GPU time was spent) and both then showed some of the strongest positive results in the entire study. This episode is documented in full in the project’s public experiment log as a cautionary methodological case study (Section 4.1).

We also swept the size of the 3,000-node official subgraph across 1,000–10,000 nodes (a $10\times$ range): all four sizes tested pass the gate ($\delta = 1.00, 1.00, 0.84, 0.81$ respectively), so the official finding is not an artefact of the specific 3,000-node choice. Separately, zero-shot evaluation of the officially trained network at input sensing radii $0.1\times$ and $5\times$ the training radius (a $50\times$ total range) both pass the gate ($\delta = 0.684$ and 0.906 respectively, $n = 30/n = 8$), and mechanism analysis shows this generalisation axis collapses under the exact same null model (`community_preserving`) as the main finding – i.e. it is not an independent second mechanism.

3.4 Boundary: where the advantage does not hold

The advantage does not generalise to: abstract, non-spatial optimisation landscapes (Sphere, Rastigrin, Ackley, Rosenbrock, Beale; $\delta \approx 0$ to -0.34); a discrete-classification reformulation of the same underlying task (octant classification, $\delta = 0.141$); closed-loop, multi-step control variants of the same task (δ falls to ≈ 0); a different species (*C. elegans*, all null models n.s.); a different individual of the same species (male *Drosophila* CNS, $\delta = 0.250$, n.s., confirmed by two independent follow-up controls); or an external gradient-free search algorithm warm-started with the network’s predictions (PSO warm-start hybrid, $\delta = 0.125$, n.s.).

Two further axes show a *reversed* advantage (real connectome performing significantly *worse* than the null): zero-shot transfer of a slope-stability-trained network to the mechanically related beam-design task without retraining ($\delta = -0.464$, $n = 30$), and robustness to moderate (15%) proportional input noise ($\delta = -0.911$, $n = 30$, confirmed independently via both ratio- and absolute-error metrics). By contrast, an initially reported reversed-advantage finding for random structural damage (edge ablation) was retracted after independent audit revealed it to be a ratio-metric artefact (the real network’s already-lower baseline error mechanically inflated a ratio-based fragility metric); recomputed on absolute error, the official 20% ablation level shows no significant difference ($\delta = -0.176$, $p = 0.246$).

Finally, the advantage evaporates under weight quantisation at every tested precision (8-, 4-, and 2-bit, all non-significant, $n = 8-30$), indicating that it depends on full floating-point numerical precision and should not be expected to transfer directly to low-bit neuromorphic hardware.

4 Discussion

Taken together, the results support a narrow but well-evidenced claim: a specific biological connectome can carry a structural computational advantage for the class of computation it plausibly evolved for (dense sensory input funnelled to a small motor output, computed in a single feed-forward pass), and this advantage is (i) acquired from remarkably little training data, (ii) robust to substantial changes in input sensing scale and subgraph size, (iii) largely but not fully attributable to modular network organisation, and (iv) fragile outside this narrow task format – to a different task class, a different species or individual, input noise, and reduced numerical precision.

This is a materially narrower and more precisely bounded claim than “biological networks are generally better computers.” It is, however, a substantially *broader* claim within its own boundary than we initially believed: what first appeared to be an idiosyncratic result on two specific engineering problems turned out, once two calibration bugs were found and fixed, to hold across six mechanically distinct physical design criteria with no exceptions.

4.1 Limitations and lessons

All official findings derive from a single female specimen’s connectome; subgraph-seed sampling ($n = 11$) supports but does not prove full generality. The **RateBrain** architecture is a biologically plausible but simplified abstraction (no spike timing, no short-term plasticity). Five additional task categories attempted in an overnight autonomous experiment batch (temporal integration, working memory, anomaly detection, chaotic time-series prediction, time-to-contact estimation) were found, on independent audit, to be invalid due to a shared infrastructure bug (a mismatch between full-graph connectivity verification and subgraph-extraction breadth-first search that can leave a single encode neuron disconnected from all decode neurons within the extracted subgraph) and yield no usable evidence either way; the root cause is documented but the shared code was deliberately left unmodified pending careful, isolated validation against the existing official results.

We highlight, as a methodological contribution in its own right, that two of the six physical-task results in Section 3.3 were initially wrong in a diagnosable, correctable way, and that both corrections were found only because an unrelated newly designed benchmark’s pre-training diagnostic numbers were suspiciously similar to an already-reported result’s diagnostics. We now treat a pre-training check of (a) the fraction of the objective contributed by each additive term at sampled points, and (b) the cosine similarity between the finite-difference teacher signal and a near-exact local gradient reference, as mandatory before any new single-shot task is trusted – several of the exciting-looking early results in this line of work (including, at different points, results in both directions) turned out on inspection to be measurement artefacts rather than genuine findings about the connectome, and only systematic self-auditing caught them.

5 Conclusion

The female *Drosophila* FlyWire connectome’s sensory-to-motor subgraph shows a robust, mechanism-partially-explained, reproducible computational advantage over degree-matched random networks, specifically on single-shot, feed-forward, continuous spatial/physical direction regression – and this advantage now extends, following correction of two calibration errors, across every physical engineering design criterion we have tested. The advantage does not indicate general-purpose computational superiority: it does not transfer across task format, species, individuals, input noise, or numerical precision. All code, raw results, and a complete, dated experiment log –

including every error found and corrected during the study – are publicly available at https://github.com/AutoPyloter/fly_op.

References

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